

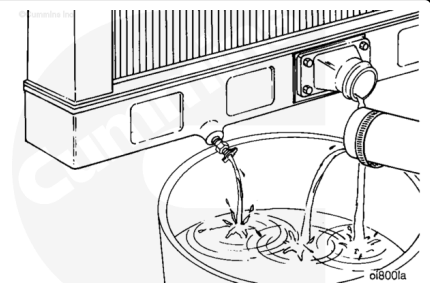
Trainings ▾

Preparatory Steps

QSK45 Engines

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

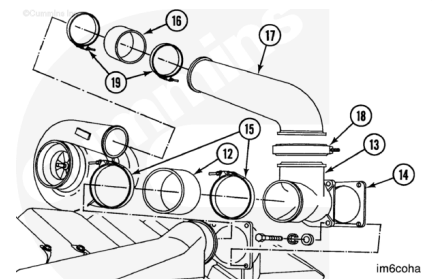


⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).

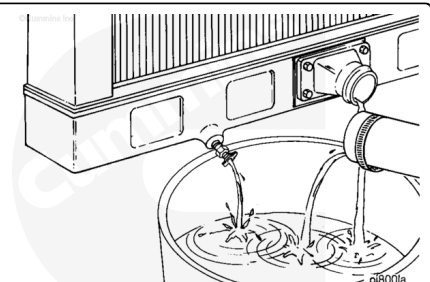
Remove the air crossover from the turbocharger and aftercooler. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>)



QSK60 Industrial and Power Generation

⚠ WARNING ⚠

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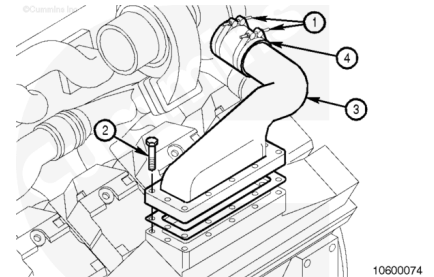
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Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left bank front and right bank rear air crossovers are different from those used on the right bank front and left bank rear, but the removal procedures are the same.

Remove the air crossover from the turbocharger and aftercooler. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>).



QSK60 Marine Applications

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left bank front and right bank rear air crossovers are different from those used on the right bank front and left bank rear, but the removal procedures are the same.

- Drain the cooling system. Refer to Procedure 008-018 (</qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html>).
- Remove the aftercooler heat shields. Refer to Procedure 010-129 (</qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html>).

- Remove the air crossover from the turbocharger and aftercooler. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>).

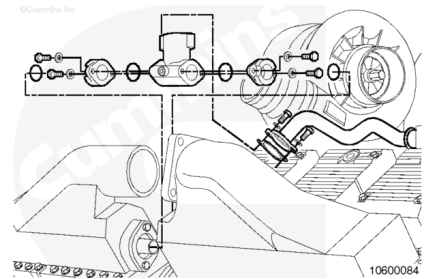
Remove

QSK45 Engines

Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

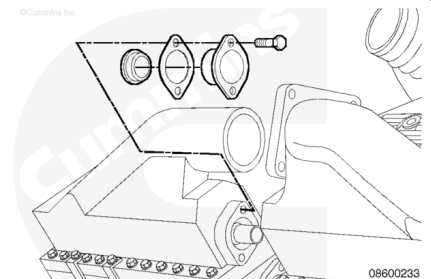
Remove the six capscrews, three covers, gaskets, and seals from each water transfer tee-connection of the two aftercooler tubes.

Remove the two tee connections from the aftercooler tubes.



Remove the capscrews, seal retainers, gaskets, and seals from each aftercooler.

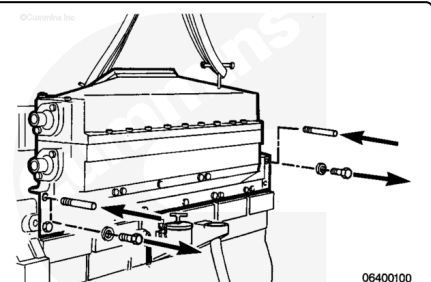
Discard the seals and gaskets.



⚠ WARNING ⚠

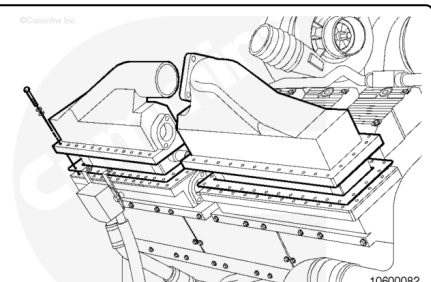
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Remove the aftercooler assembly. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>).



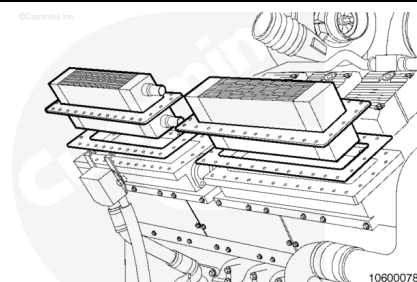
Remove the aftercooler cover capscrews, cover, and gasket.

Discard the seals and gaskets.



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This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



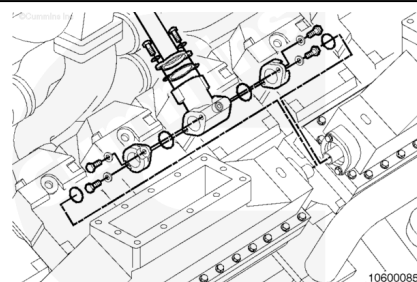
Remove the aftercooler element and gaskets.

Discard the gaskets.

QSK60 Engines

Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

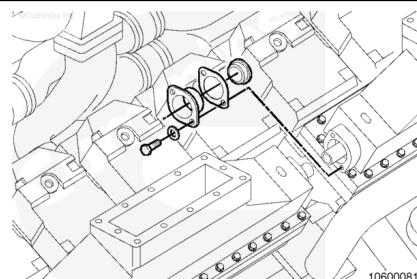
Remove the six capscrews, three covers, gaskets, and seals from each water transfer tee connection of the two aftercooler tubes.



Remove the two water transfer tee connections from the aftercooler tubes.

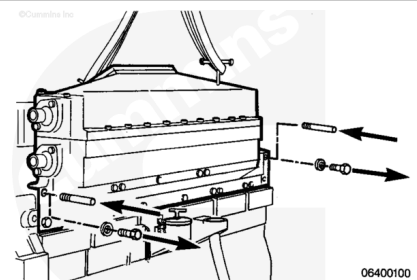
Remove the capscrews, seal retainers, gaskets, and seals from each aftercooler.

Discard the seals and gaskets.



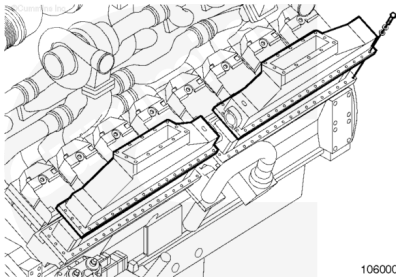
⚠ WARNING ⚠

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Remove the aftercooler assembly. Refer to Procedure 010-002 (</qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html>).

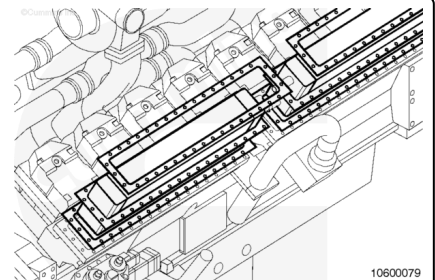
Remove the aftercooler cover capscrews and cover.



10600083

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This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



10600079

Remove the aftercooler element and gaskets.

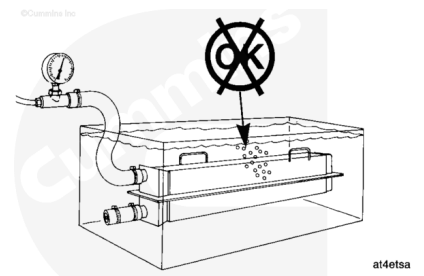
Discard the gaskets.

Pressure Test

Use a hose plugged with a pipe plug and attach it to one water tube as shown.

Attach an air line and gauge to the remaining tube.

Apply air pressure and check for leaks.



at4etsa

Measurements

	kpa	psi
Air Pressure	415	60

Heating the tank water to 50°C [122°F] improves the test results.

Check for aftercooler element leaks.

Do **not** attempt to repair a leaking element. The element **must** be replaced.

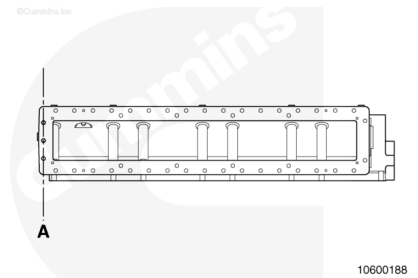
Machine

QSK60 Engines

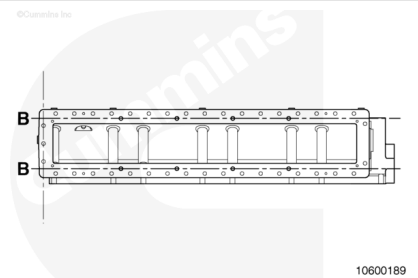
Note : This procedure only shows one side of the air intake manifold and aftercooler cover, the opposite side is a mirror image of the first side. The dimensions and reference lines are all identical.

The early design air intake manifold and cover require the dowel pins to be moved in order to fit the new aftercooler elements.

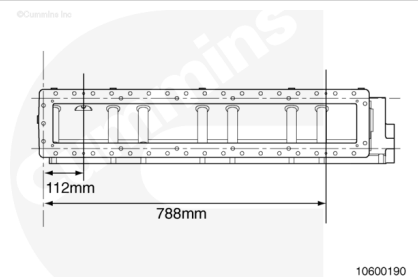
Draw a line through the middle of the end three capscrew holes of the air intake manifold as indicated by line A.



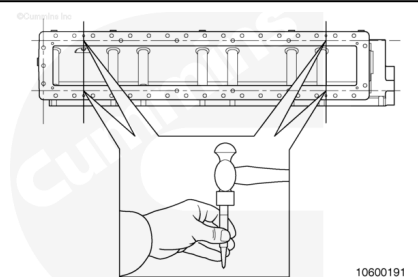
Draw two lines through the middle of the aftercooler core mounting capscrew holes as indicated by line B.



Using line A as the reference line, measure from A along both line B's and mark two marks on each line. One at 112 mm [4.41 in] from line A, and one at 788 mm [31.02 in] from line A.



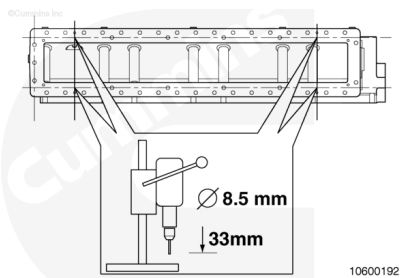
Center punch the four holes.



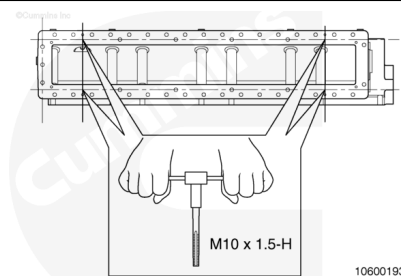
Attach the air intake manifold to a vertical drill.

Drill the four holes using an 8.5 mm drill to a depth of 33 mm [1.3 in].

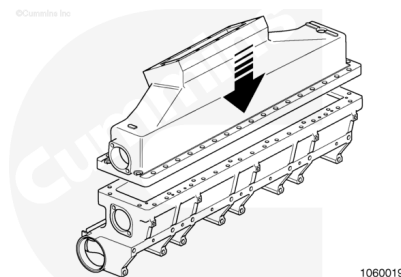
Clean the holes and remove the debris.



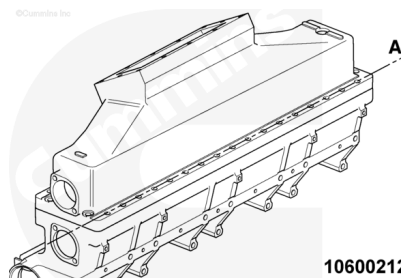
Tap the four holes using a M10 X 1.5-H tap.



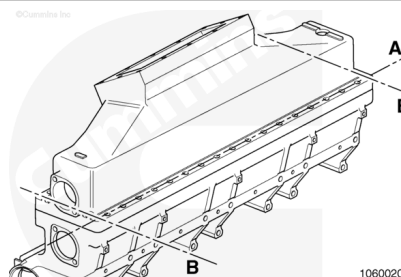
Assemble the air intake manifold and aftercooler cover without the aftercooler core.



Draw a line through the center of the capscrew holes shown in line A.

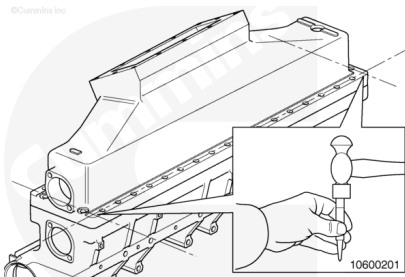


Draw a line through the center of the capscrew holes shown in lines A and B.



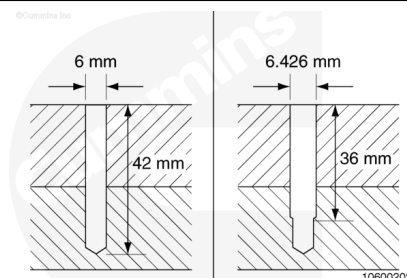
Center punch the cover where the lines intersect.

Attach the assembly to a vertical drill.



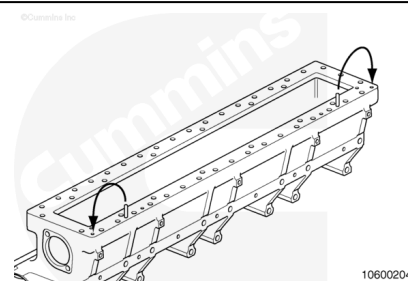
Drill two holes where marked.

Use a 6 mm drill and drill to a depth of 42 mm [1.65 in] to enlarge the holes using a 6.426 mm drill to a depth of 36 mm [1.417 in].

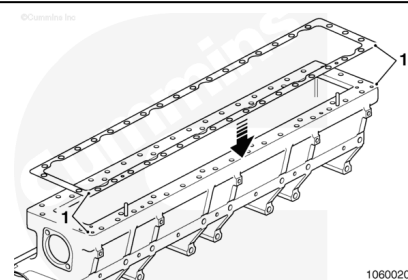


Disassemble the aftercooler cover and air intake manifold.

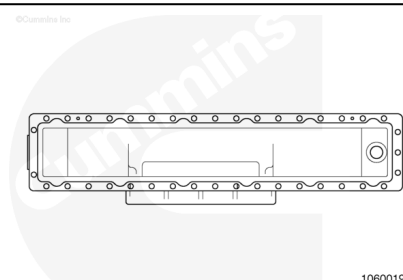
Remove the dowel pins from their old location and fit them in the two new holes.



Place the new gasket over the air intake manifold and add two extra holes in the gasket for the dowel pins.

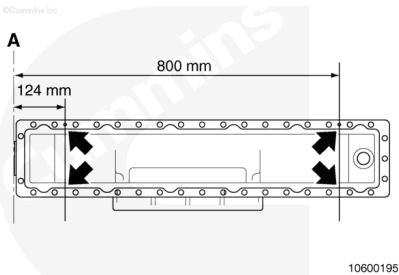


The old aftercooler covers, Part Number 4001329 and 4001330, need four extra slots added to provide clearance for the four new aftercooler mounting capscrews.



Mark the cover using the dimensions shown.

Reference line A is the edge of the cover where the aftercooler core water outlet protrudes.

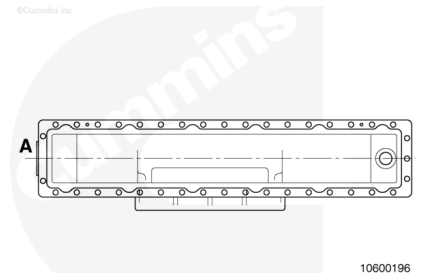


Attach the marked aftercooler cover to a milling machine.

Align the center of the slot drill with the marks in the aftercooler cover and the center line A.

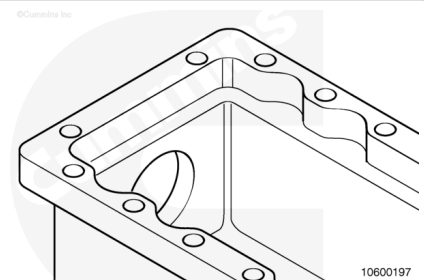
Move the slot/flat faced drill away from the center line 66 mm [2.59 in].

Use a 35 mm [1.38 in] slot drill to machine four new slots to a depth of 20 mm [0.79 in].

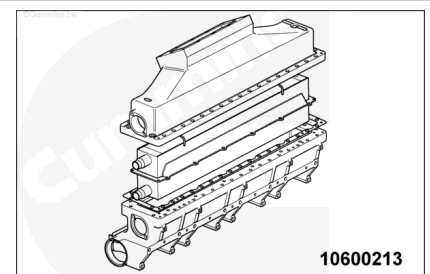


The new slots should be the same as the existing 12 slots.

Thoroughly clean the cover and remove any debris.



Assemble the modified parts.



The old aftercooler core **must not** be used if a crack in the mounting flange is found. See Figures 1 and 2.

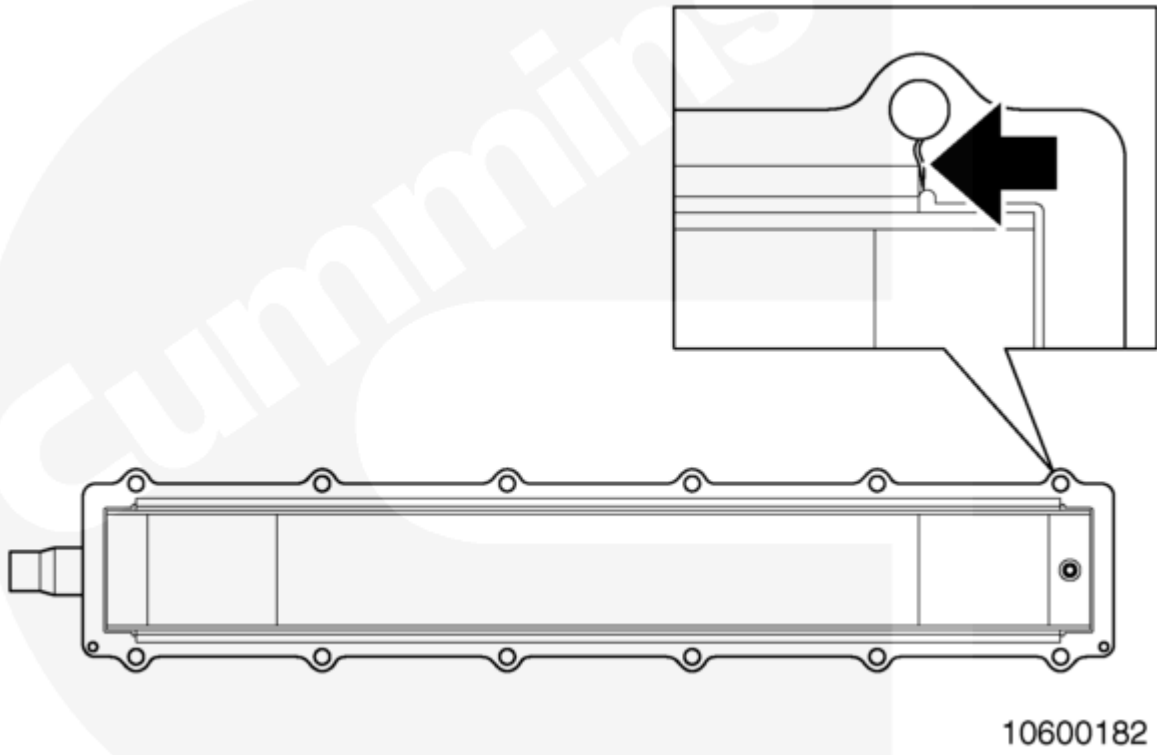


Figure 1

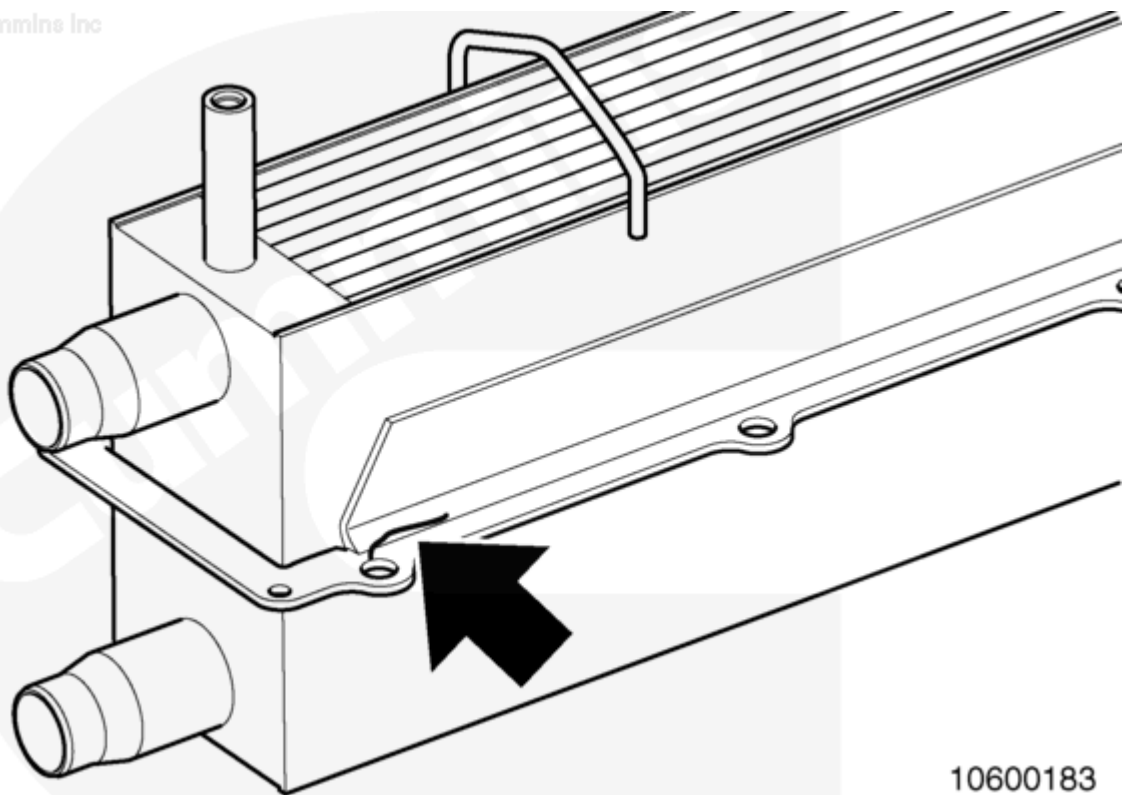
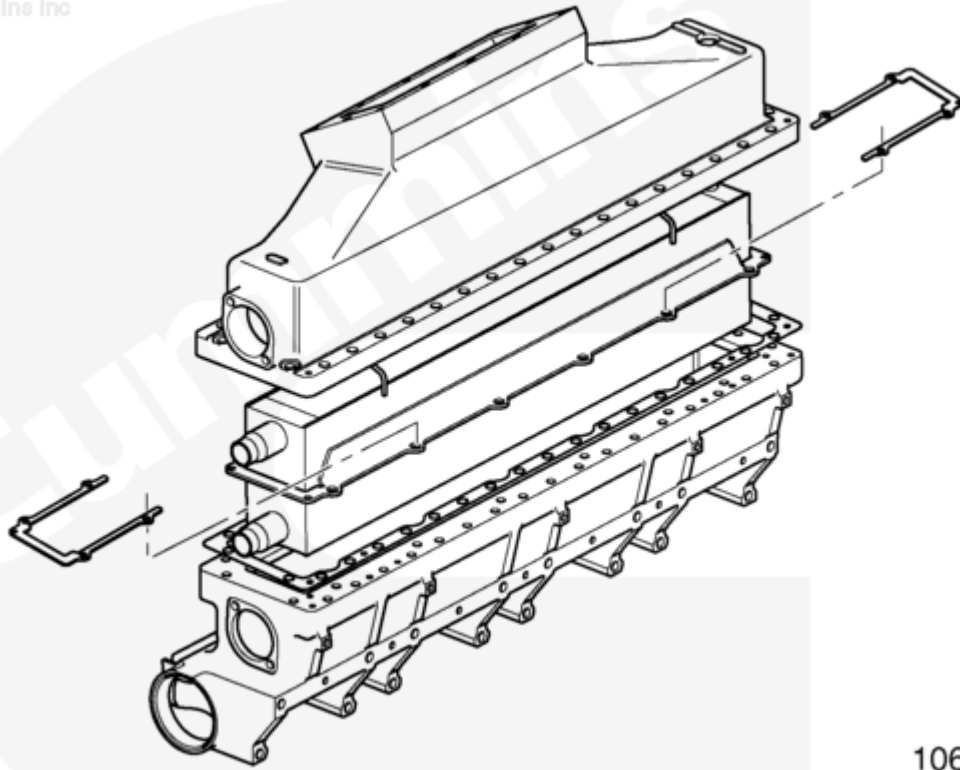


Figure 2

If cracks are **not** found on the aftercooler core, recondition it using the U-shaped clamping plate, Part Number 4066244, is recommended. The new plate will help lower the assembly stress and increase the strength of the aftercooler core flange.



10600208

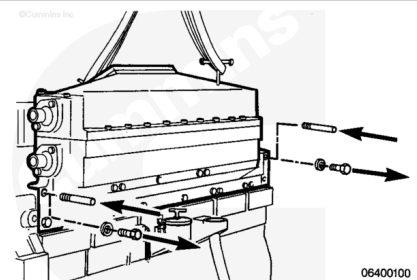
Install

QSK45 Engines

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

Install the aftercooler housing. Refer to Procedure 010-002 ([/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html](http://qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)).

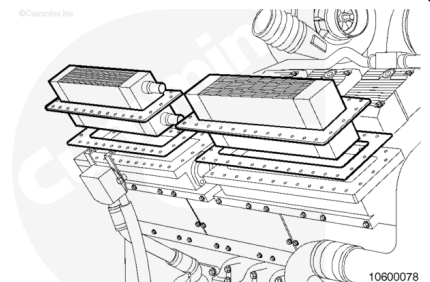


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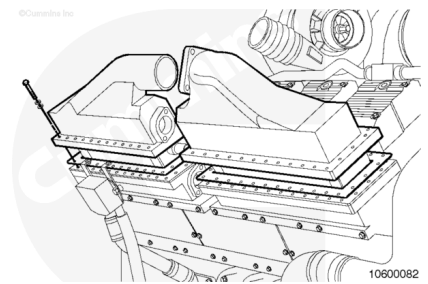
Make sure the aftercooler element supports are positioned correctly.

Install the aftercooler elements and gaskets.



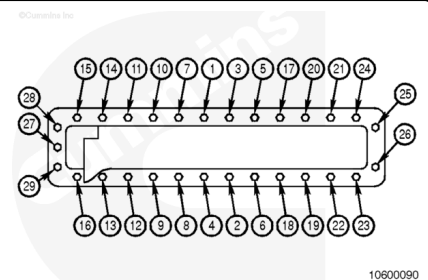
Coat the threads of all of the aftercooler cover capscrews with anti-seize lubricant, Part Number 3824759.

Position the aftercooler cover on the aftercooler and install the 29 capscrews.



Tighten the capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft-lb]



Lubricate the seal with clean engine oil.

Install the seal on the retainer.

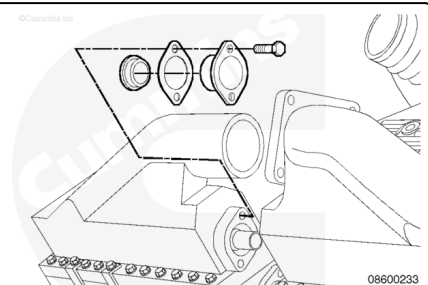
Lubricate the outside diameter of the water bosses on the aftercooler element with Lubriplate™ 105.

Slide the gasket on the water bosses.

Slide the seal and the retainer on the tube.

Install and tighten the capscrews.

Torque Value: 10 n•m [89 in-lb]



Note : Procedures for one side of the engine are shown. Procedures for the other side are similar.

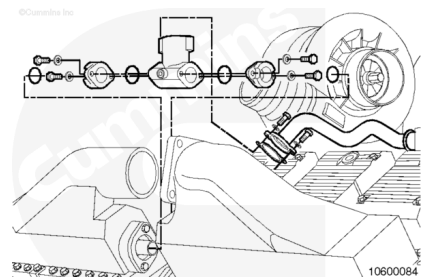
Position the water transfer tee connection covers, gaskets, and seals on the four aftercooler tubes.

Position the two tee connections on the four aftercooler tubes.

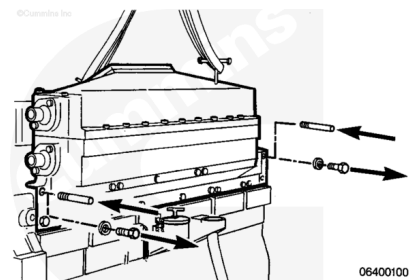
Install three covers, gaskets, and seals on the two tee connections of the four aftercooler tubes with six capscrews.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]



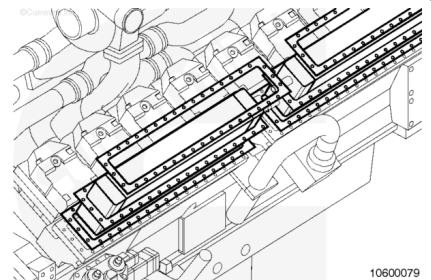
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Install the aftercooler housing. Refer to Procedure 010-002 (/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html).

⚠ WARNING ⚠

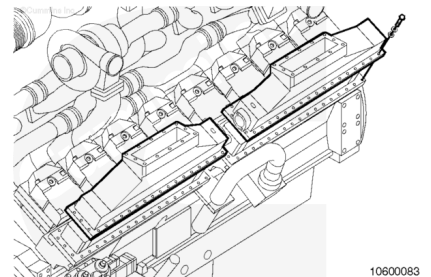
This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



Make sure the aftercooler supports are positioned correctly.
Install the aftercooler elements and new gaskets.

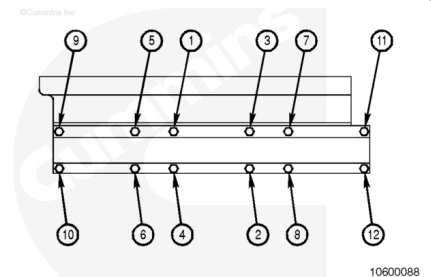
Coat the threads of all aftercooler cover capscrews with anti-seize lubricant, Part Number 3824759.

Position the aftercooler cover on the aftercooler and install the capscrews.



Tighten the aftercooler cover capscrews in the sequence shown.

Torque Value: 45 n•m [33 ft•lb]



Lubricate the seal with Lubriplate™ 105.

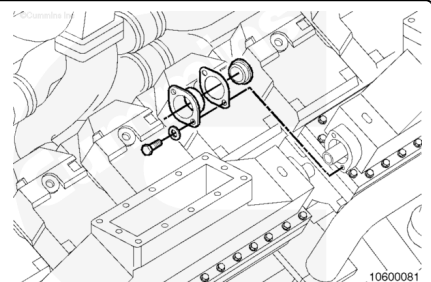
Install the seal on the retainer.

Lubricate the outside diameter of the water bosses on the aftercooler element with clean engine oil.

Slide the gasket on the water bosses.

Slide the seal and the retainer on the tube.

Install and tighten the capscrews.



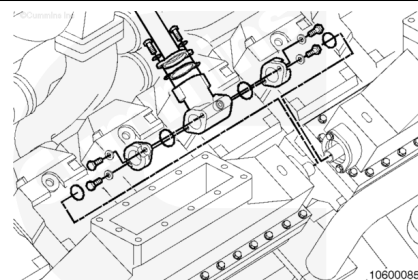
Torque Value: 10 n•m [89 in-lb]

Install the water transfer tee connection covers on the aftercooler outlet and inlet tubes.

Install the water transfer tee connection between the aftercoolers, and place the water supply tube in the water transfer connection.

Tighten the capscrews on the end covers of the water transfer tee connection.

Torque Value: 10 n•m [89 in-lb]



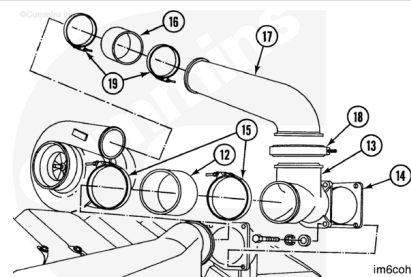
Finishing Steps

QSK45 Engines

Install the air crossover from the turbocharger and aftercooler.

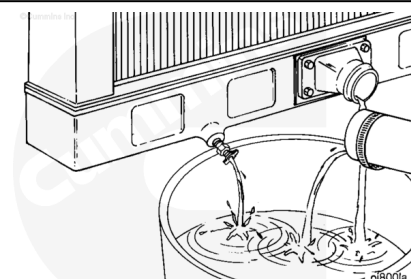
Refer to Procedure 010-002

([/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html](http://qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html))



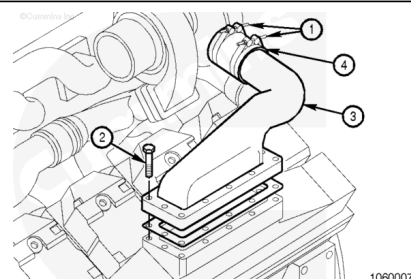
Fill the cooling system. Refer to Procedure 008-018

([/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html](http://qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)).



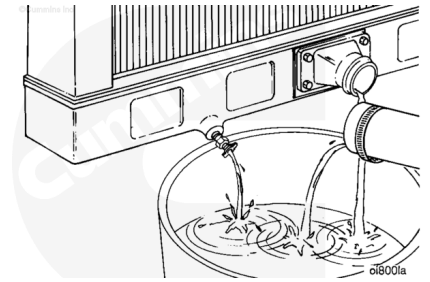
QSK60 Industrial and Power Generation

Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left bank front and right bank rear air crossovers are different from those used on the right bank front and left bank rear, but the removal procedures are the same.



Install the air crossover from the turbocharger and aftercooler.
Refer to Procedure 010-002
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html).

Fill the cooling system. Refer to Procedure 008-018
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html).



QSK60 Marine Applications

Note : The QSK60 engine contains an individual air crossover for each aftercooler. The configuration of the left bank front and right bank rear air crossovers are different from those used on the right bank front and left bank rear, but the removal procedures are the same.



- Install the air crossover from the turbocharger and aftercooler. Refer to Procedure 010-002
(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html).
- Install the aftercooler heat shields. Refer to Procedure 010-129 (/qs3/pubsys2/xml/en/procedures/56/56-010-129-tr.html).
- Fill the cooling system. Refer to Procedure 008-018
(/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html).